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online compiler and debugger for c/c++

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main.c

```
stdio.h>
string.h>

s Modulo-2 division via bitwise XOR logic
vide(char data[], char generator[], char remainder[]) {
taLen = strlen(data);
nLen = strlen(generator);
emp[100];

y the initial segment of data into a temporary working array
y(temp, data, genLen);
enLen] = '\0';

inter = genLen;
(pointer <= dataLen) {
 If the leading bit is '1', perform XOR with the generator polynomial
 (temp[0] == '1') {
  for (int i = 1; i < genLen; i++) {
      temp[i - 1] = (temp[i] == generator[i]) ? '0' : '1';
  }
else {
  // If the leading bit is '0', XOR with all '0's (slides the frame forward)
  for (int i = 1; i < genLen; i++) {
      temp[i - 1] = (temp[i] == '0') ? '0' : '1';
  }

 Fetch the next bit from the data stream
 (pointer < dataLen) {
  temp[genLen - 1] = data[pointer];

inter++;

 final remainder length is always (Generator Length - 1)
y(remainder, temp, genLen - 1);
der[genLen - 1] = '\0';

 {
ata[50], generator[20], appendedData[70], remainder[20], codeWord[70], recei
 dataLen, genLen;

Sender Side Code Execution
("--- SENDER SIDE ---\n");
("Enter Raw Data Bits: ");
"%s", data);
("Enter Generator Polynomial Bits: ");
"%s", generator);

n = strlen(data);
 = strlen(generator);

d appendedData with original data bits
(appendedData, data);

end (N - 1) zeros to the data bits stream
 = 0; i < genLen - 1; i++) {
pendedData[dataLen + i] = '0';

edData[dataLen + genLen - 1] = '\0';

pute the CRC Checksum
ide(appendedData, generator, remainder);
("Generated CRC Checksum: %s\n", remainder);

struct final codeword: original data + calculated remainder
(codeWord, data);
(codeWord, remainder);
("Codeword Transmitted over Network: %s\n\n", codeWord);

Receiver Side Verification Simulation
("--- RECEIVER SIDE ---\n");
("Enter Received Codeword Stream: ");
```

input

```
--- SENDER SIDE ---
Enter Raw Data Bits: 100100
Enter Generator Polynomial Bits: 001
Generated CRC Checksum: 11
Codeword Transmitted over Network: 10010011

--- RECEIVER SIDE ---
Enter Received Codeword Stream: 10010011
Resulting Remainder: 00
Status: SUCCESS! Data received flawlessly with zero errors.


...Program finished with exit code 0
Press ENTER to exit console.
```